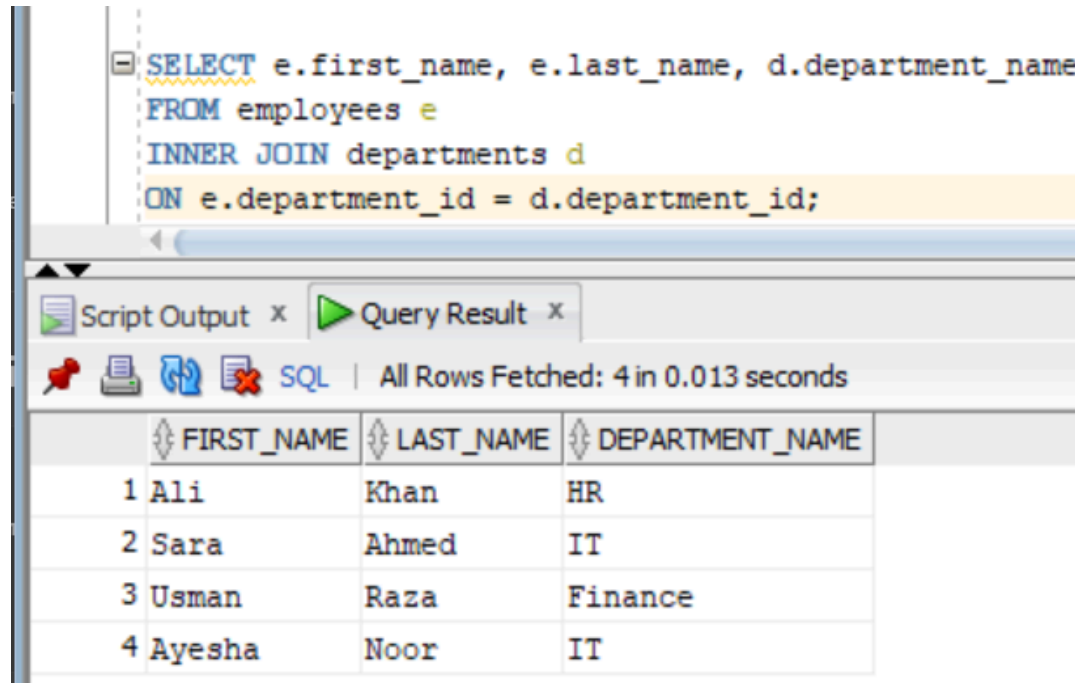


DB Lab06 Submission

Name: Yash Raj

Roll Number: 24k-0737

Task 01:



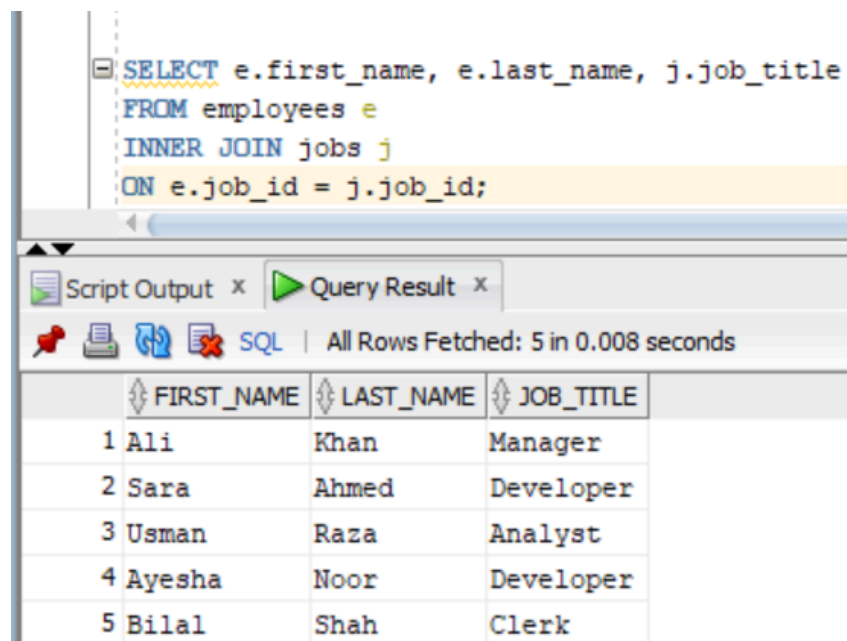
The screenshot shows a SQL query window with the following query:

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
INNER JOIN departments d
ON e.department_id = d.department_id;
```

Below the query window, the 'Query Result' tab is active, showing the results of the query. The status bar indicates 'All Rows Fetched: 4 in 0.013 seconds'.

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Ali	Khan	HR
2	Sara	Ahmed	IT
3	Usman	Raza	Finance
4	Ayesha	Noor	IT

Task 02:



The screenshot shows a SQL query window with the following query:

```
SELECT e.first_name, e.last_name, j.job_title
FROM employees e
INNER JOIN jobs j
ON e.job_id = j.job_id;
```

Below the query window, the 'Query Result' tab is active, showing the results of the query. The status bar indicates 'All Rows Fetched: 5 in 0.008 seconds'.

	FIRST_NAME	LAST_NAME	JOB_TITLE
1	Ali	Khan	Manager
2	Sara	Ahmed	Developer
3	Usman	Raza	Analyst
4	Ayesha	Noor	Developer
5	Bilal	Shah	Clerk

Task 03:

```
SELECT e.first_name, e.last_name, d.location_id
FROM employees e
JOIN departments d
USING (department_id);
```

Script Output x Query Result x

SQL | All Rows Fetched: 4 in 0.006 seconds

	FIRST_NAME	LAST_NAME	LOCATION_ID
1	Ali	Khan	1
2	Sara	Ahmed	2
3	Usman	Raza	3
4	Ayesha	Noor	2

Task 04:

```
SELECT first_name, last_name, department_name
FROM employees
NATURAL JOIN departments;
```

Script Output x Query Result x

SQL | All Rows Fetched: 4 in 0.013 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Ali	Khan	HR
2	Sara	Ahmed	IT
3	Usman	Raza	Finance
4	Ayesha	Noor	IT

Task 05:

```
SELECT e.first_name AS employee_name, m.first_name AS manager_name
FROM employees e
JOIN employees m
ON e.manager_id = m.employee_id;
```

Script Output x Query Result x

SQL | All Rows Fetched: 4 in 0.01 seconds

	EMPLOYEE_NAME	MANAGER_NAME
1	Sara	Ali
2	Usman	Ali
3	Ayesha	Sara
4	Bilal	Usman

Task 06:

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
LEFT JOIN departments d
ON e.department_id = d.department_id;
```

Script Output x Query Result x

SQL | All Rows Fetched: 5 in 0.011 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Ali	Khan	HR
2	Sara	Ahmed	IT
3	Ayesha	Noor	IT
4	Usman	Raza	Finance
5	Bilal	Shah	(null)

Task 07:

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
RIGHT JOIN departments d
ON e.department_id = d.department_id;
```

Script Output x Query Result x

SQL | All Rows Fetched: 5 in 0.006 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Ali	Khan	HR
2	Sara	Ahmed	IT
3	Usman	Raza	Finance
4	Ayesha	Noor	IT
5	(null)	(null)	Marketing

Task 08:

```
SELECT e.first_name, d.department_name
FROM employees e
CROSS JOIN departments d;
```

Script Output x Query Result x

SQL | All Rows Fetched: 20 in 0.009 second

	FIRST_NAME	DEPARTMENT_NAME
12	Sara	Finance
13	Usman	Finance
14	Ayesha	Finance
15	Bilal	Finance
16	Ali	Marketing
17	Sara	Marketing
18	Usman	Marketing
19	Ayesha	Marketing
20	Bilal	Marketing

Task 09:

```
SELECT employee_id FROM employees
UNION
SELECT employee_id FROM job_history;
```

Script Output x Query Result x

SQL | All Rows Fetched: 5 in 0.015 seconds

	EMPLOYEE_ID
1	101
2	102
3	103
4	104
5	105

Task 10:

```
SELECT employee_id FROM employees
INTERSECT
SELECT employee_id FROM job_history;
```

Script Output x Query Result x

SQL | All Rows Fetched: 2 in 0.011 seconds

	EMPLOYEE_ID
1	103
2	102

Task 11:

```
SELECT e.first_name, e.last_name, d.department_name, l.city
FROM employees e
JOIN departments d ON e.department_id = d.department_id
JOIN locations l ON d.location_id = l.location_id;
```

Script Output x Query Result x

SQL | All Rows Fetched: 4 in 0.009 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME	CITY
1	Ali	Khan	HR	Karachi
2	Sara	Ahmed	IT	Lahore
3	Usman	Raza	Finance	Islamabad
4	Ayesha	Noor	IT	Lahore

Task 12:

```
CREATE OR REPLACE VIEW emp_dept_view AS
SELECT e.employee_id, e.first_name, e.last_name, d.department_name
FROM employees e
JOIN departments d ON e.department_id = d.department_id;
```

Script Output x Query Result x

Task completed in 0.063 seconds

1 row inserted.

Commit complete.

View EMP_DEPT_VIEW created.

Task 13:

```
CREATE OR REPLACE VIEW high_salary_emp AS
SELECT *
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

Script Output x Query Result x

Task completed in 0.057 seconds

Commit complete.

View EMP_DEPT_VIEW created.

View HIGH_SALARY_EMP created.

Task 14:

```
SELECT employee_id, first_name, last_name
FROM employees
WHERE employee_id NOT IN (SELECT employee_id FROM job_history);
```

Script Output x Query Result x

All Rows Fetched: 3 in 0.015 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME
1	105	Bilal	Shah
2	104	Ayesha	Noor
3	101	Ali	Khan